D03 · Shop Floor Execution & MES

01 · MES system selection & implementation

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

MES (Manufacturing Execution System) is the operational software layer between the ERP (which sees the plant as inventory and financial flows) and the floor (which sees machines, operators and physical product). It orchestrates work orders, tracks production in real time, enforces process discipline, and captures the shop floor data that the rest of the enterprise needs.

2 WHY IT MATTERS

The MES decision is the most consequential operational technology choice a plant makes — and the one most often delegated to IT. **Without an effective MES, the plant runs on tribal knowledge, lagging spreadsheets, and the supervisor's memory**. The dominant failure pattern is documented in the MESA International model: plants that implement MES as an IT project (rather than an operations transformation) generate Gartner-tracked failure rates of 30–50% within five years **(MESA International, 2024 — industry body)**. The asymmetry is structural — **MES decisions are committed for 7–12 years but their value depends on daily operator adoption**, and adoption is determined by usability long before the technical specifications matter.

3 HOW IT IS DONE

1. **Define the operational outcomes the MES must deliver — not the features.** A typical specification lists 200 features; a useful specification lists 5 outcomes (OEE visibility within 15 seconds, work order release without paper, lot traceability in <30 seconds, etc.). Vendors meet feature lists; outcomes force them to demonstrate.
2. **Choose architecture before vendor — discrete, process, hybrid, ISA-95 layer.** Discrete MES (Rockwell FactoryTalk, Siemens Opcenter) and process MES (AVEVA, Emerson DeltaV) are not interchangeable; ISA-95 Level 3 vs. Level 4 boundaries determine where MES ends and ERP begins. Getting this wrong creates integration debt that compounds for years.
3. **Score vendors against operator UX, not screenshots.** Sit three operators in front of three vendor demos doing the same realistic task — work order start, material consumption, defect report. The vendor that produces working operators in 20 minutes beats the vendor with prettier dashboards every time.
4. **Pilot on the hardest line, not the easiest.** The temptation is to pilot on the cleanest, most stable line for an easy win. The result is an MES that works in showroom conditions and breaks on contact with reality. Pilot on the line with the most product variants, the most setups, and the messiest data — that is where the system will live.
5. **Design the ERP-MES integration before signing the MES contract.** 60% of MES implementation cost lives in integration. Define the exact data flows (work order release, confirmation, material movement, quality results) and the latency requirements (real-time vs. batch) before the vendor estimates the project.
6. **Plan for the 18-month adoption curve, not the go-live event.** Go-live is week zero; the value comes from sustained usage that takes 12–18 months to embed. Budget for ongoing operator training, supervisor coaching, and the iteration cycles to fix the things the pilot did not catch.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> An automotive Tier 1 with 4 plants and 28 production lines evaluates Rockwell, Siemens, GE Digital, and a niche specialist. The specification document is rewritten from 312 features to 6 outcomes. The operator UX test on Day 2 of the vendor demos eliminates 2 of the 4 vendors — operators couldn't complete a work order start in under 5 minutes. The selected platform pilots on the highest-mix line (847 SKUs, 23 setups/day). At month 6, **OEE on the pilot line improves 11.4 points**, with the gain coming from minor stop reduction (real-time visibility eliminated reactive debugging). Total implementation cost: **$3.8M across 4 plants over 22 months**; payback in 16 months from OEE gains alone.

IN PRACTICE

Operations leads, IT supports

The plant manager and operations director own the outcomes; IT owns the technical integration. Inverting this is the most common organizational failure — the resulting system integrates well with the data warehouse and is hated by every operator on the floor.

Real operator veto on the selection committee

Line supervisors and senior operators sit on the committee with real vote weight — not advisory roles. Advisory roles get overruled at the moment that matters, and by the time that is discovered, the contract is signed.

ISA-95 boundary discipline

Level 3 is MES, Level 4 is ERP, Levels 1–2 are sensors and PLCs **(ISA-95 / IEC 62264 — international standard)**. Crossing those boundaries informally creates point-to-point integrations that nobody wants to own three years later.

Budget the 18-month adoption curve, not just go-live

Go-live is week zero; the value comes from sustained usage that takes 12–18 months to embed. Budget for ongoing operator training, supervisor coaching, and iteration cycles to fix the things the pilot did not catch.

WHAT YOU WOULD USE

MES PLATFORMS

Siemens Opcenter Execution — Strong fit for discrete manufacturing, especially in automotive and aerospace; deep PLM integration with Teamcenter; the choice when the plant already has substantial Siemens automation footprint **(Siemens — vendor)**.

Rockwell FactoryTalk ProductionCentre — The dominant choice in North American discrete manufacturing, with tight integration to Rockwell PLCs (Allen-Bradley installed base); strong in CPG, food and beverage, and automotive Tier 1s **(Rockwell — vendor)**.

AVEVA MES (formerly Wonderware) — Process industry strength — chemicals, pharma, food and beverage — with native integration to the AVEVA System Platform **(AVEVA — vendor)**.

Cloud-native modular MES (Tulip, Solumina, MESbox) — The newer cohort — operator-facing, low-code, faster to deploy, more constrained at scale. The right choice for plants with high variability and lighter integration needs **(Tulip — vendor; Solumina — vendor)**.

4 SO WHAT

The MES that operators will not use is a server, not a system. Specify outcomes not features, choose architecture before vendor, score on operator UX, pilot on the hardest line, design the ERP integration upfront, and budget the 18-month adoption curve. Operations leads the selection; IT supports it. Done right, the MES is the floor's nervous system; done wrong, it is a $3M reporting tool.

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